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End Semester Examination December, 2024

Name of the Course: ECE

Name of the Paper: Digital Electronics

Time: 3 Hours

Semester: IIIrd

Paper Code: TEC 302

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Solve the following expression using K-map $Y = \sum m(1,3,7,9,13,15,21,25,31) + d(0,2,5,10,17,23,27)$	CO1
(b)	Implement the following expression using 8:1 multiplexer and 3:8 decoder $Y = \sum m(1,3,6,8,10,12,15)$	
(c)	Design a circuit of 4 inputs which gives high output only when parity of input combination is odd.	
Q2	(20 marks)	
(a)	Design a BCD to Exs-3 code converter.	CO2
(b)	What is the difference between i) PLA and PAL ii) Ring counter and Johnson counter iii) Fan In and Fan Out.	
(c)	Implement the circuit of Full adder using PLA.	
Q3	(20 marks)	
(a)	Explain race around condition. How it can be eliminated using Master Slave Flip flop.	CO3
(b)	Explain working of universal shift register with suitable diagram.	
(c)	What is the difference between Synchronous and Asynchronous counter? Design a synchronous counter for sequence 1→2→4→6→1 using T Flip Flop.	
Q4	(20 marks)	
(a)	What are Registers? Explain 4 –bit SIPO shift register with SR Flip flop.	CO4, CO6
(b)	A Synchronous sequential logic circuit with input x is described by the following function $T_A = x \text{ AND } Q_B$ $T_B = x \text{ XOR } Q_A$ Analyze the circuit accordingly	
(c)	Design a 2-bit counter with external input p. The counter counts up when input is 0 otherwise it remain in its state.	
Q5	(20 marks)	
(a)	Explain features of TTL family. Draw and explain TTL NAND Gate.	CO5, CO6
(b)	Explain operation of CMOS NOR Gate.	
(c)	Classify different logic families also explain their characteristics.	